

- 8 (a) (i) $2mu$ A1
- (ii) $2L / u$ A1
- (iii) force = $N \times \text{change in momentum} / \text{time} = N 2mu / (2L / u)$
 $= Nmu^2 / L$ A1
- (iv) pressure = force / area = $(Nmu^2 / L) / L^2$
 $= Nmu^2 / L^3$ A1
- (b) (i) Molecules has component of velocity in three directions. B1
- or $c^2 = u_x^2 + u_y^2 + u_z^2$
- or $\langle c^2 \rangle = \langle u_x^2 \rangle + \langle u_y^2 \rangle + \langle u_z^2 \rangle$
- Since the molecules are in random motion, on average,
- $\langle u_x^2 \rangle = \langle u_y^2 \rangle = \langle u_z^2 \rangle$ B1
- Thus
- $\langle c^2 \rangle = 3\langle u_x^2 \rangle$
- $\langle u_x^2 \rangle = \frac{1}{3} \langle c^2 \rangle$ B1
- From (a) (iv), $p = Nmu^2 / L^3$
or $pV = Nmu^2$
- $pV = \frac{1}{3}Nm \langle c^2 \rangle$ A0
- (ii) $pV = NkT$ C1
- $NkT = \frac{1}{3}Nm \langle c^2 \rangle$
- multiply $\frac{3}{2}$ to both sides: $\frac{1}{2}m \langle c^2 \rangle = \frac{3}{2}kT$ and $\frac{1}{2}m \langle c^2 \rangle = E_K$ M1
- so $E_K = \frac{3}{2}kT$ A0
- (c) $\frac{1}{2} \times 3.34 \times 10^{-27} \times \langle c^2 \rangle = \frac{3}{2} \times 1.38 \times 10^{-23} \times (25 + 273)$ C1
- r.m.s. speed = $1.9 \times 10^3 \text{ m s}^{-1}$ A1

- (d) internal energy = ΣE_K of molecules + ΣE_P of molecules **B1**
no forces between molecules, so potential energy of molecules is zero **B1**
- (e) (i) increase in internal energy = Q + work done **B1**
 constant volume so no work done **B1**
- (ii) thermal energy per unit mass to cause a unit change in temperature **B1**
- (iii) $c = Q / Nm\Delta T$ **C1**
 $= [N \times (3/2)k\Delta T] / (Nm\Delta T)$ **M1**
 $= 3k / 2m$ **A0**
- (f) As the gas expands, it does work against the atmosphere/external pressure **B1**
 for same temperature rise, more thermal energy needed, so larger specific heat capacity **B1**